

Survival Differences Across PAM50 Subtypes in Breast Cancer

Background

Breast cancer remains the leading cause of cancer-related deaths among women worldwide. Despite advances in early detection and breast cancer treatment, it remains a major public health challenge due to its heterogeneity. Tumors have varying molecular subtypes with distinct clinical outcomes and treatment responses. Molecular profiling has improved the ability to stratify patients beyond traditional clinical factors. One such tool is the PAM50 classifier. PAM50 is a 50-gene signature used to categorize tumors into intrinsic subtypes including Luminal A, Luminal B, HER2-enriched, Basal-like, and normal-like [1]. Alongside the five subtypes, claudin-low is included as an additional subtype under this classification system. While PAM50 is widely used to guide treatment decisions, less is known about how its prognostic impact may evolve over time. Understanding these time-varying risks could help tailor monitoring strategies and long-term treatment planning.

A collaborative study conducted with METABRIC (Molecular Taxonomy of Breast Cancer International Consortium), analyzed an extensive dataset of approximately 2,000 primary breast tumors, including detailed information on DNA copy number changes and gene expression profiles, along with long-term clinical follow-up data [2]. This paper utilizes the METABRIC dataset to investigate whether PAM50 molecular subtypes are significant prognostic indicators of overall survival in breast cancer, independent of other clinical features. I hypothesize that breast cancer patients classified into different PAM50 molecular subtypes exhibit significant differences in overall survival, even after adjusting for age, tumor size, and histologic grade. The analysis will implement a multivariable Cox proportional hazards model as the basis for this investigation.

Methods

Prior to the analysis, missing data in crucial variables such as survival time and status were addressed by removing those observations, reducing the dataset size from 2,509 to 1,870 observations. The PAM50 Not Classified (PAM50NC) group was also excluded due to insufficient events and lack of relevance.

A multivariable Cox proportional hazards model was then constructed to assess the association between PAM50 subtype and overall survival. As PAM50 is the primary predictor of interest, it was converted to a categorical variable with Luminal A as the reference category, as it is considered the most favorable prognosis (longest survival). Tumor size was included as a continuous covariate with age at diagnosis and histologic grade. Age at diagnosis was grouped into bins: <40, 40-49, 50-59, 60-69, and 70+, and converted to a factor along with neoplasm histologic grade which has levels 1, 2, and 3.

The proportional hazards assumption was tested using Schoenfeld residuals, which indicated significant violations, particularly for age and histologic grade. To account for this, the model was stratified on the age groups and histologic grade and incorporated time-dependent effects by splitting time into three intervals: 0-40 months, 40-80 months, and 80-170 months. Patients with

follow-up beyond 170 months were censored. Interaction terms between the PAM50 subtypes and time intervals were generated to allow hazard ratios to vary across the three periods. This threshold was based on the Schoenfeld residual patterns, which revealed a violation of the proportional hazards assumption before 170 months and stabilization thereafter. This approach balances model validity with statistical power and is consistent with methods used to address time-dependent effects in survival analysis [3].

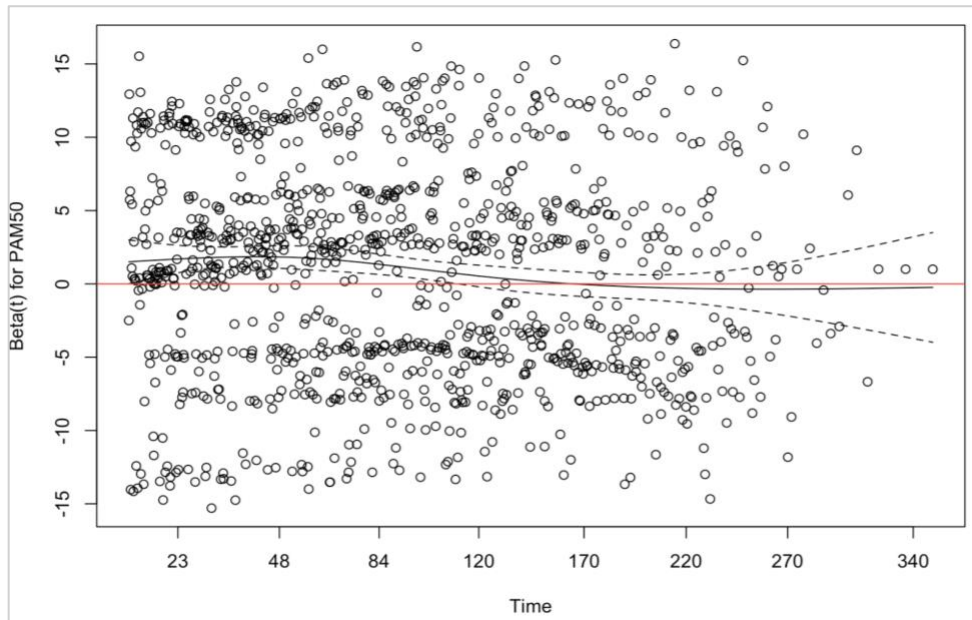


Figure 1. Schoenfeld residual plot for PAM50 after stratification of age and tumor grade

The linearity of continuous covariates in the Cox model was also assessed using Martingale residual plots. The residuals plot (Fig. 2) shows that the LOWESS line starts below 0, curves upward, and then plateaus at higher tumor sizes, suggesting that the linearity assumption for tumor size does not hold.

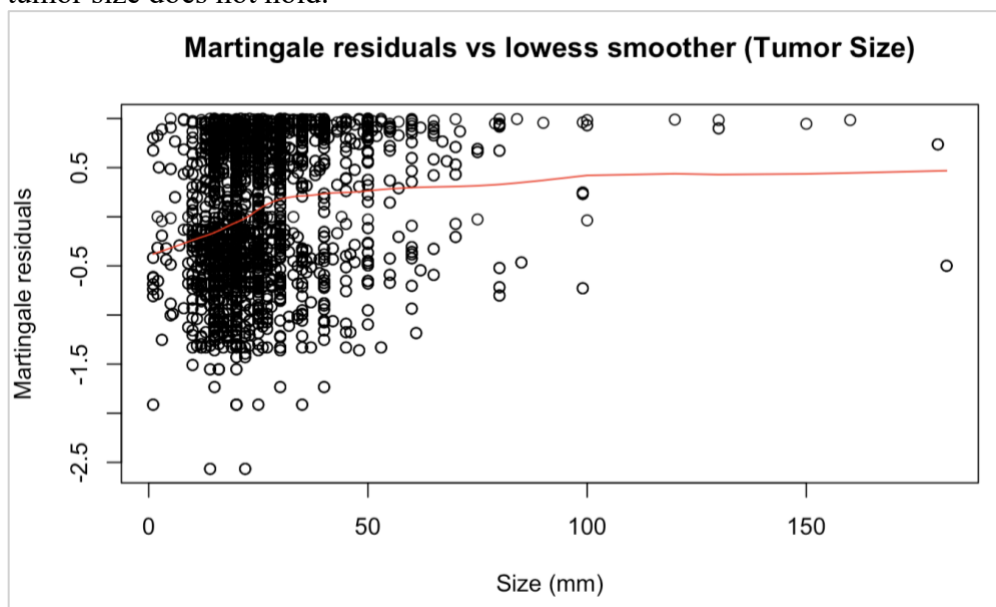


Figure 2. Martingale residuals plot for tumor size

To address this, the Cox model was refitted using a natural cubic spline for tumor size with 4 degrees of freedom. A likelihood ratio test comparing the linear and spline models showed that the spline significantly improved model fit (Chi-squared = 22.83, df = 3, $p < 0.0001$), confirming that tumor size has a non-linear association with the hazard of death. Thus, the final model used to assess survival outcomes included PAM50 subtypes as time-dependent covariates across three time intervals (0-40, 40-80, and 80-170 months), tumor size modeled using a natural cubic spline with 4 degrees of freedom, and stratification by age group and histologic grade to account for non-proportional hazards.

Results

The Kaplan-Meier curve (Fig. 3) shows that Luminal A type tumors have the best survival as it has the highest curve throughout, which is consistent with their favorable prognosis. Claudin-low also shows relatively good survival for much of the follow-up period but has steeper drops as time increases. The Luminal B and normal-like subtypes indicate intermediate survival as the lines lie between the best and worst prognostic groups. Basal-like and HER2 subtypes present the worst survival with the earliest and steepest declines over time. Additionally, the log-rank test p-value is $p < 0.0001$, indicating a statistically significant difference in survival curves across PAM50 subtypes. The Kaplan-Meier curves show unadjusted survival by PAM50 subtype over the full range of follow up data. These do not account for covariates, stratification, or time-capping used in the Cox model.

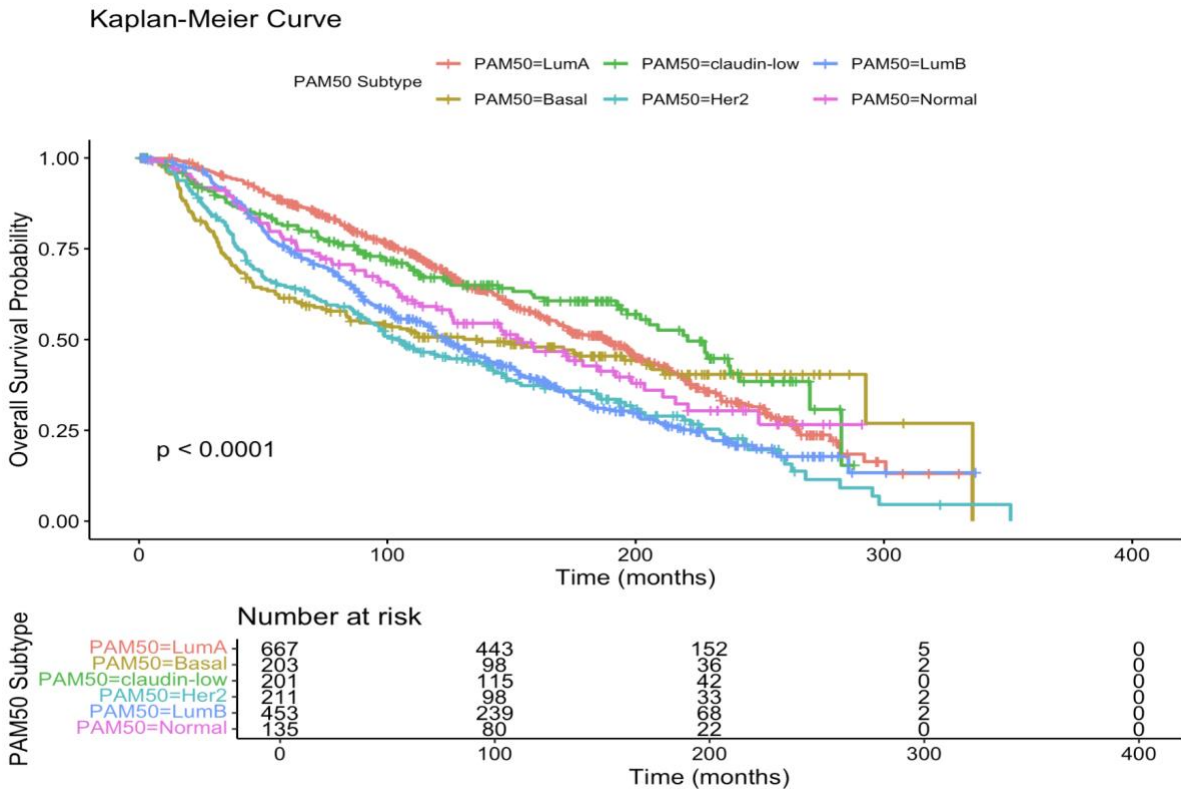


Figure 3. Kaplan-Meier curve of overall survival across PAM50 subtypes

Table 1 summarizes the hazard ratios from the time-dependent Cox model with follow-up capped at 170 months. The model includes time-varying effects for each PAM50 subtype across three intervals: 0-40, 40-80, and 80-170 months.

Variable	HR (exp(coef))	95% CI	p-value
LumB_0_40	1.46	0.96 – 2.22	0.079
LumB_40_80	1.62	1.17 – 2.24	0.004
LumB_80_170	1.18	0.92 – 1.53	0.197
Basal_0_40	3.87	2.49 – 6.01	0.000
Basal_40_80	0.94	0.56 – 1.57	0.804
Basal_80_170	0.59	0.35 – 1	0.050
Her2_0_40	3.26	2.11 – 5.02	0.000
Her2_40_80	1.35	0.87 – 2.09	0.177
Her2_80_170	1.11	0.77 – 1.61	0.568
Normal_0_40	2.35	1.34 – 4.14	0.003
Normal_40_80	1.68	1.02 – 2.76	0.040
Normal_80_170	1.20	0.79 – 1.84	0.393
Claudin_low_0_40	1.98	1.19 – 3.27	0.008
Claudin_low_40_80	0.81	0.48 – 1.38	0.437
Claudin_low_80_170	0.75	0.47 – 1.17	0.203
ns(Tumor.Size, df = 4)1	1.73	0.85 – 3.52	0.133
ns(Tumor.Size, df = 4)2	5.13	2.79 – 9.41	0.000
ns(Tumor.Size, df = 4)3	3.96	0.8 – 19.58	0.092
ns(Tumor.Size, df = 4)4	2.80	0.8 – 9.8	0.108

Compared to Luminal A, Basal-like tumors had the highest hazard of death during the first 40 months (HR = 3.87, 95% CI: 2.49-6.01, $p < 0.001$), but this effect diminished in later intervals. In fact, Basal-like tumors were associated with a significantly lower risk of death from 80-170 months (HR = 0.59, 95% CI: 0.35-1.00, $p = 0.050$). HER2-enriched tumors similarly showed elevated early risk (HR = 3.26, 95% CI: 2.11-5.02, $p < 0.001$), but their effects attenuated and were non-significant beyond 40 months.

Luminal B tumors had 62% higher hazards during 40-80 months (HR = 1.62, 95% CI: 1.17-2.24, $p = 0.004$), with weaker and non-significant associations in other intervals compared to Luminal A. The normal-like subtype was associated with significantly elevated risk in both the 0-40 month (HR = 2.35, 95% CI: 1.34-4.14, $p = 0.003$) and 40-80 month intervals (HR = 1.68, 95% CI: 1.02-2.76, $p = 0.041$), but not after 80 months.

Claudin-low tumors were significantly associated with higher mortality early in follow-up (HR = 1.98, 95% CI: 1.19-3.27, $p = 0.008$), but not in later time periods. The effect weakened and was

non-significant between 40-80 months (HR = 0.81, 95% CI: 0.48-1.38, $p = 0.437$) reflecting a 19% lower risk and 80-170 months (HR = 0.75, 95% CI: 0.47-1.17, $p = 0.203$) reflecting a 25% lower hazard, suggesting that Claudin-low tumors may carry early risk that declines over time.

While not all spline components were individually significant, the overall association of tumor size and mortality risk was supported by a likelihood ratio test comparing models with and without the spline term (Chi-squared = 69.4, $p < 0.001$), confirming that tumor size is a significant predictor of overall survival. The nonlinear association between tumor size and survival is further illustrated by the plot of the estimated hazard ratio using a natural cubic spline (Fig 4.). Predictions were generated for the most common patient stratum (age 60-69, histologic grade 3). The curve shows a slight increase in hazard for tumors below 50 mm, with a steeper rise for larger sizes, indicating that the risk of death accelerates with increasing tumor burden. This supports the use of splines over a linear term to capture the nonlinear effect of tumor size.

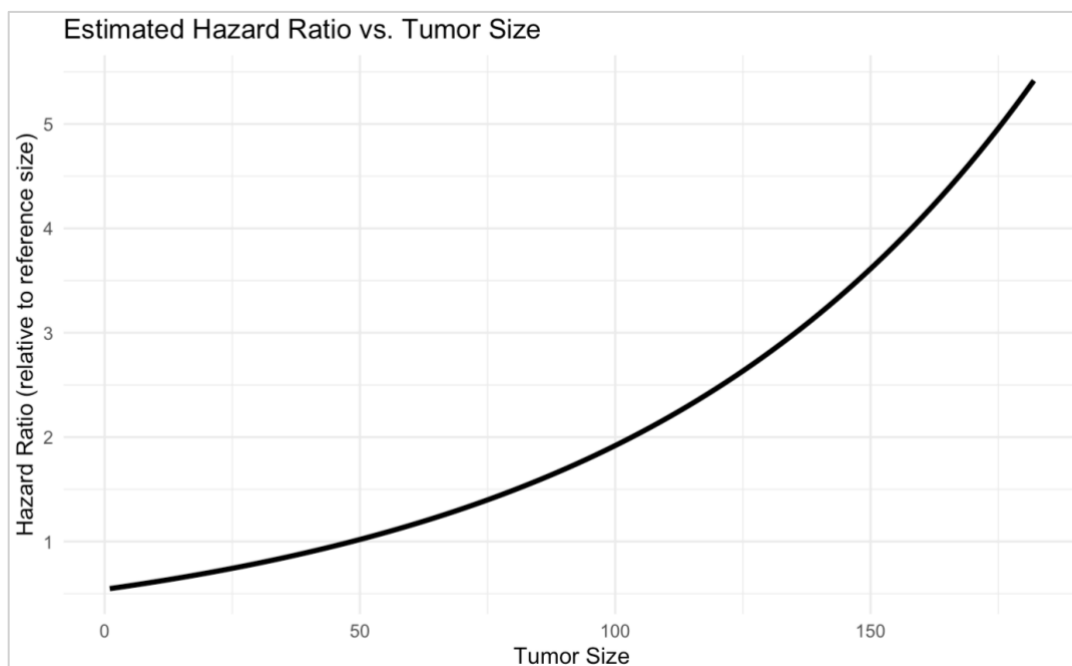


Figure 4. Nonlinear effect of tumor size on hazard of death, modeled with natural cubic splines (df = 4). Estimates shown for the most common stratum (age 60-69, grade 3)

Overall, the final Cox model demonstrated strong global significance (likelihood ratio test $p < 2e-16$), and the time-dependent modeling approach offered a more nuanced understanding of how subtype-specific risk evolves over time. Incorporating a flexible spline term for tumor size further enhanced model fit and interpretability.

Discussion

This study hypothesized that breast cancer patients classified into different PAM50 molecular subtypes exhibit significant differences in overall survival, even after adjusting for clinical features such as age, tumor size, and histologic grade. The findings partly support this hypothesis. Basal-like and HER2-enriched tumors had substantially higher hazards of death early in follow-up (0-40 months), reinforcing their aggressive behavior and the necessity for close observation and early intervention. In contrast, the elevated risk for Luminal B tumors appeared

later, particularly between 40-80 months, indicating a more prolonged pattern of recurrence or mortality. These results reflect the known biological characteristics of Basal and HER2 tumors' rapid progression and lower responsiveness to endocrine therapy, while Luminal B tumors may exhibit more persistent risk over time [4].

This analysis also revealed a significant early risk associated with the Claudin-low subtype (HR = 1.98, $p = 0.008$), which decreased and became non-significant in later follow-up. This suggests that Claudin-low tumors, though initially high-risk, may not sustain an elevated mortality hazard over the long term. Normal-like tumors showed a similar time-dependent pattern, with elevated hazard observed early and mid follow-up, but attenuated effects after 80 months.

Tumor size was modeled using a natural cubic spline with 4 degrees of freedom to capture potential non-linearity. It was a significant, nonlinear predictor of overall mortality (likelihood ratio test $p < 0.001$), supporting its continued relevance in prognostic models. While smaller tumors were associated with relatively stable risk, hazard increased steeply beyond 50 mm, indicating a strong association between larger tumor size and poorer survival.

Capping follow-up at 170 months was based on the Schoenfeld residual plot for PAM50, which showed proportional hazards violations in earlier periods that stabilized after approximately 170 months. This cut-off allowed the model to reflect temporal variation in subtype risk while reducing distortion from late follow-up durations. However, although the overall sample size was relatively large, dividing time into multiple intervals and stratifying by age and histologic grade reduced the number of events per group, potentially limiting statistical power, particularly for some PAM50 subtypes during later follow-up periods. For example, the loss of statistical significance for Basal-like, HER2-enriched, and Claudin-low subtypes beyond 40 months may be partially explained by low event counts and right-censoring, which increase uncertainty in hazard ratio estimates. Another limitation is that although the time-varying Cox model addressed violations of the proportional hazards assumption, using manually defined intervals (0-40, 40-80, 80-170 months) is subjective and may not fully capture non-linear hazard trajectories.

It is important to note that while stratification helped account for non-proportional hazards, residual confounding may still exist. The goal of this analysis was to evaluate the association between the prognostic value of molecular subtype with survival independent of treatment decisions. Including treatment variables, which are often determined by subtype, could obscure the relationship between subtype and outcome. Thus, treatment variables (e.g. chemotherapy, hormone therapy, radiotherapy) were intentionally excluded from the model to avoid over-adjustment, as treatment is often determined by subtype and could mediate its effect on survival. However, this can also be a limitation. Omitting these variables from the final Cox model may have introduced confounding, since variations in treatment across subtypes could influence outcomes. Future studies should adjust for treatment to better isolate the prognostic effect of subtype.

Conclusion

This analysis demonstrates that the impact of PAM50 molecular subtypes on breast cancer survival is time-dependent. Basal-like, HER2-enriched and Claudin-low tumors are associated with significantly elevated mortality risk early in follow-up, while Luminal B shows increased

hazard later. These hazard profiles support the continued use of PAM50 in guiding clinical decision-making and highlight the importance of flexible modeling approaches that accommodate time-varying effects. Future research should incorporate treatment data, recurrence-free survival endpoints, and larger cohorts to validate these findings and improve precision.

References

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